

ZARA Dataset Analysis

1- Data Cleaning & Preparation:

- Checked **data types** for each column.
- Handled **missing values** – none found after cleaning.
- Converted columns from string to numeric where necessary (e.g., Sales Volume, Price).
- Checked for **duplicates** and removed if any.
- Verified **categorical consistency** for columns like Promotion, Seasonal, Section, Material, and Origin.

Outcome: Clean, consistent dataset ready for analysis.

2- Univariate Analysis:

2.1- Price

```
count    20252.000000
mean      41.949061
std       23.380960
min       12.000000
25%       23.950000
50%       35.950000
75%       53.950000
max       134.990000
Name: price, dtype: float64
```



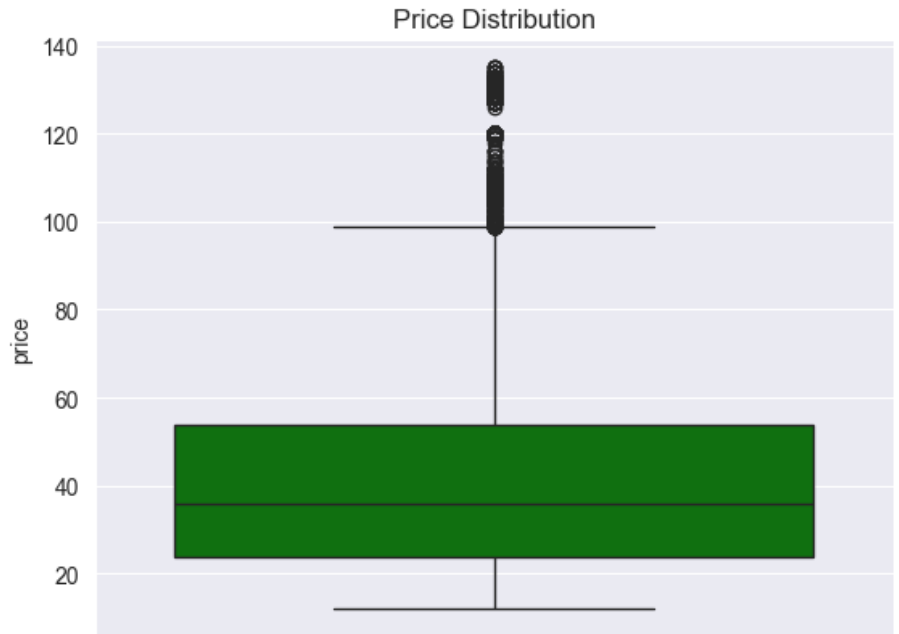
-Most products are priced between 20 and 40, indicating that Zara mainly focuses on mid-range price

There are a small number of products with higher prices, reaching 134

The distribution is skewed towards mid-range prices

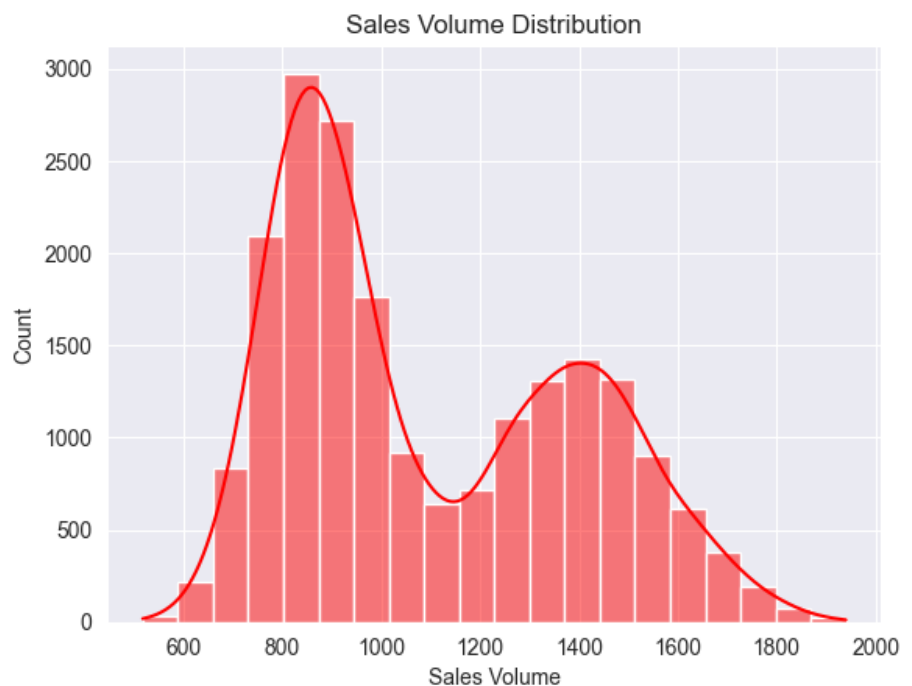
-The boxplot shows that most product prices are concentrated between 24 and 54, with a median around 36.

There are a few higher-priced outliers, indicating the presence of premium products



2.2- Sales Volume:

```
count    20252.000000
mean      1097.400454
std       298.234609
min       518.000000
25%       849.000000
50%       990.000000
75%      1364.250000
max      1940.000000
Name: Sales Volume, dtype: float64
```



- Most products sell between 800 and 1400 units. A few products sell much more, which shows the top-performing items

Most products sell mean between 849 and 1364 units, with a median of 990. A few products sell much more, indicating top-performing items



2.3-Clothing:

-All products in the dataset belong to the "clothing" category

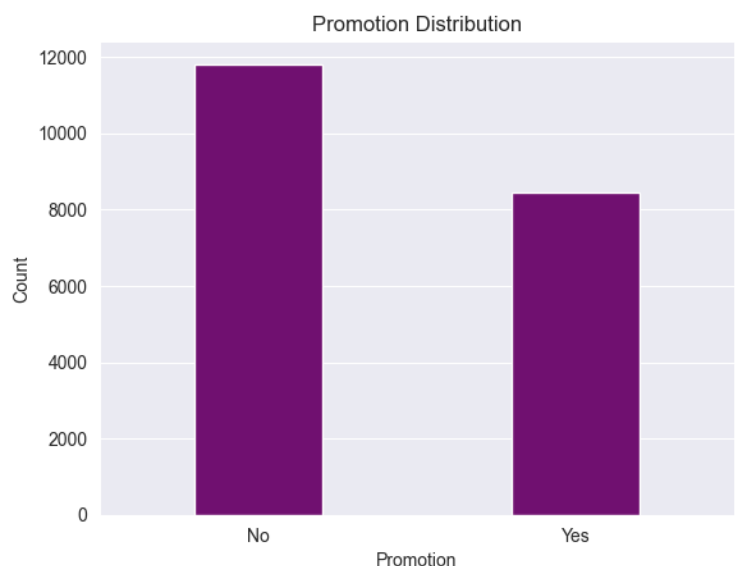
indicating that Zara's dataset focuses exclusively on apparel. No other product categories are represented.

2.4-Promotion:

```
Promotion
No      11812
Yes      8440
Name: count, dtype: int64
```

Most products are not under promotion, indicating that Zara's catalog is largely sold at regular prices.

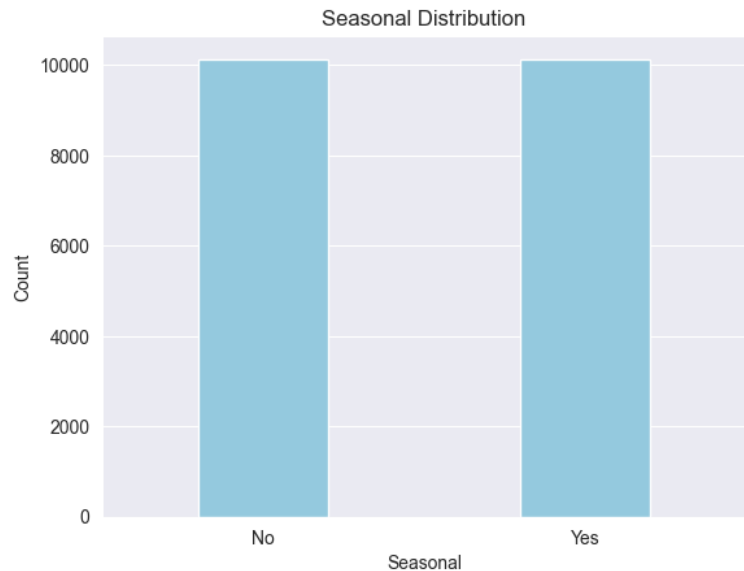
Only a small portion of products is offered under promotional campaigns



2.5-Seasonal:

```
Seasonal
No      10136
Yes     10116
Name: count, dtype: int64
```

The dataset shows a nearly equal distribution between seasonal and non-seasonal products, with 10116 seasonal items and 10136 non-seasonal items. This indicates Zara maintains a balanced catalog, offering both year-round and season-specific products.

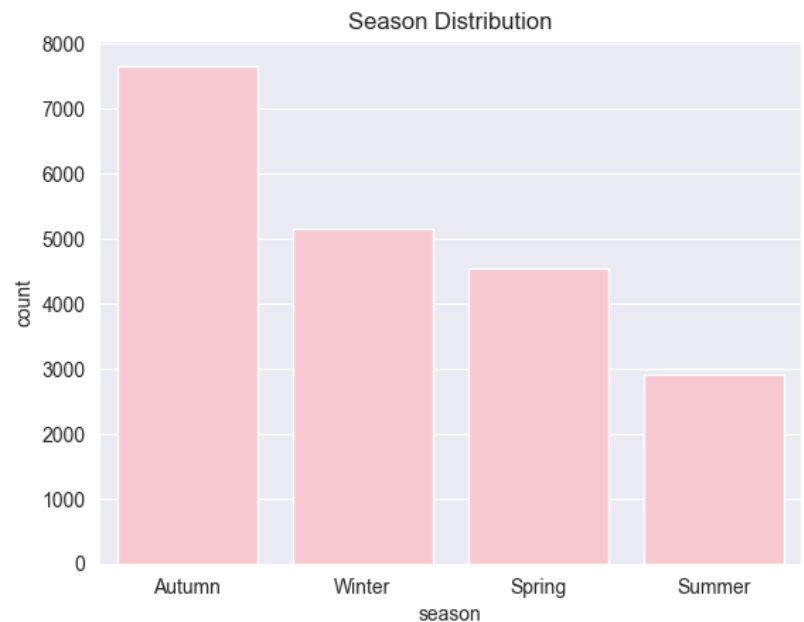


2.6-Season:

```
season
Autumn   7665
Winter   5144
Spring   4537
Summer   2906
Name: count, dtype: int64
```

The distribution of products by season shows that most items are designed for Autumn (7,665) and Winter (5,144), indicating a stronger focus on cooler seasons.

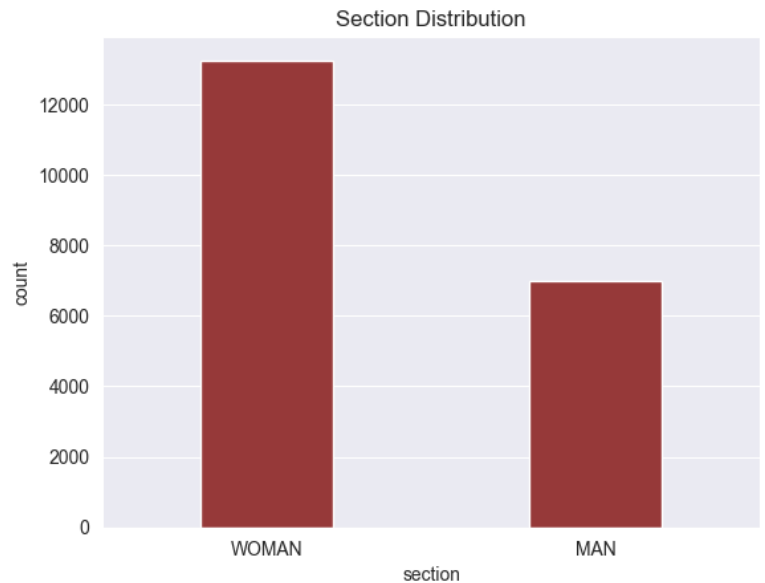
Spring (4,537) and Summer (2,906) products are fewer, reflecting seasonal inventory planning and market demand.



2.7- Section:

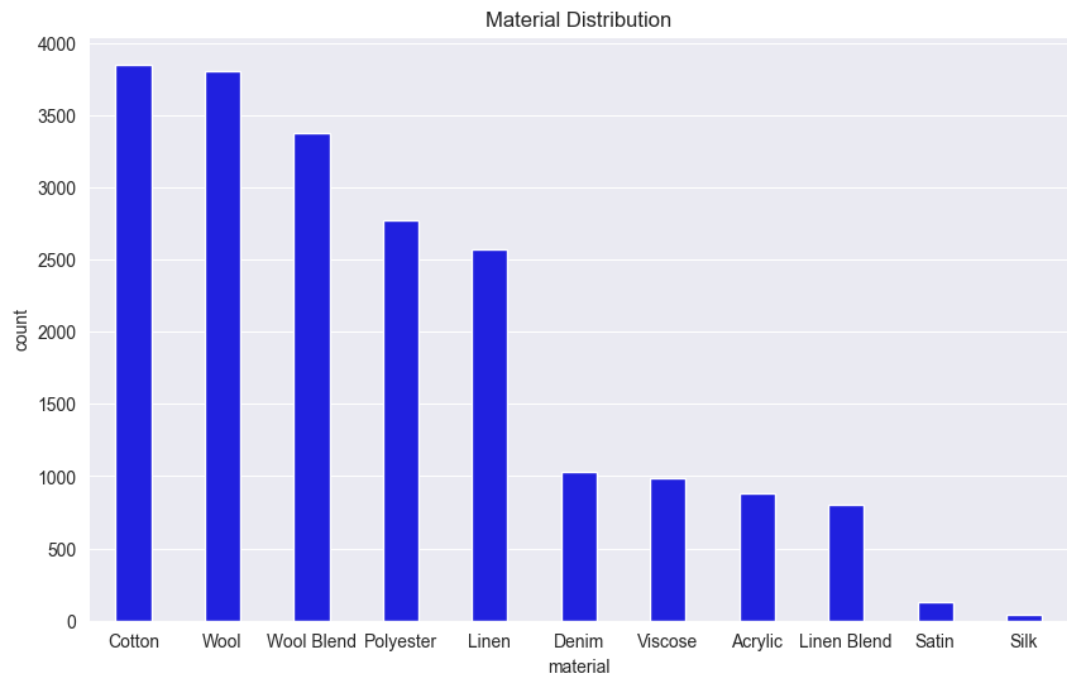
```
section
WOMAN    13254
MAN       6998
Name: count, dtype: int64
```

Most products are for Women, and Men sections having fewer items, showing Zara's focus on the Women category.



2.8- Material:

```
material
Cotton      3851
Wool        3805
Wool Blend  3373
Polyester   2775
Linen       2573
Denim       1027
Viscose     990
Acrylic     881
Linen Blend 807
Satin       132
Silk        38
Name: count, dtype: int64
```



The product distribution by material shows that Cotton, Wool, and Wool Blend dominate the catalog, reflecting Zara's focus on popular everyday fabrics.

Polyester and Linen are also significant, while premium fabrics like Satin and Silk are rare, likely reserved for special items.

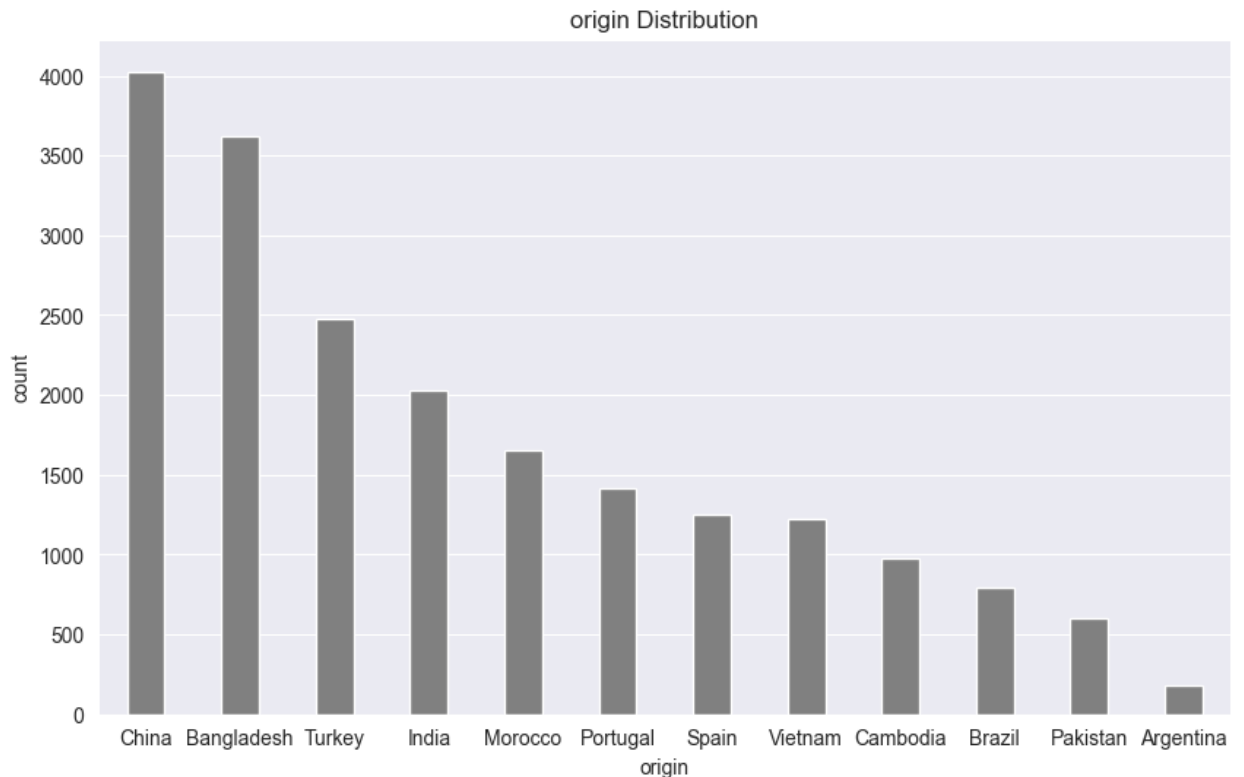
2.9- Origin:

```
origin
China      4026
Bangladesh 3617
Turkey     2475
India       2033
Morocco     1653
Portugal    1420
Spain       1248
Vietnam     1220
Cambodia    981
Brazil       795
Pakistan    605
Argentina   179
Name: count, dtype: int64
```

Most products are manufactured in China, Bangladesh, Turkey, and India, indicating Zara's reliance on large-scale production in Asia.

European countries like Portugal and Spain contribute a smaller portion,

likely representing premium or specialized items, while South American origins like Brazil and Argentina are minimal.



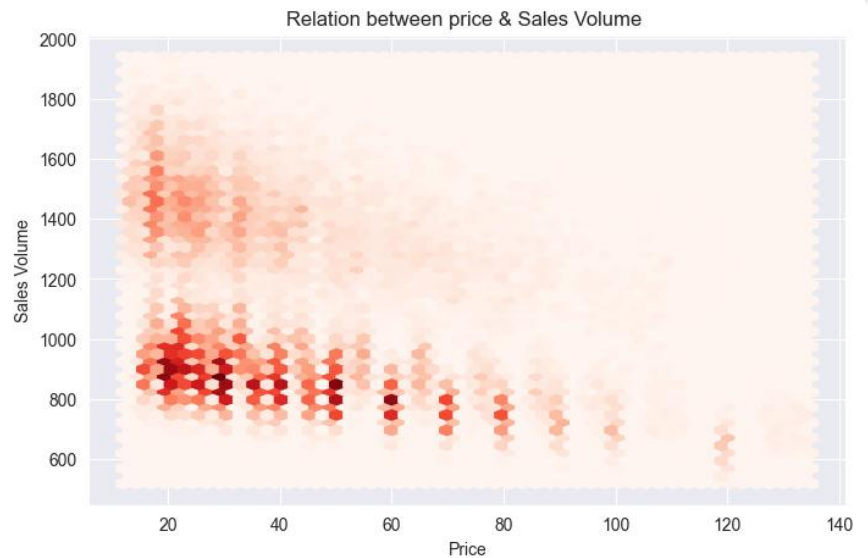
3-Bivariate Multivariate Analysis:

3.1- Price vs Sales volume:

There is a weak negative correlation between price and sales volume, indicating that higher prices slightly reduce sales.

The density plot highlights that most products cluster in the lower price range, with higher sales volumes. Very few products have high prices, and their sales volumes are generally low. This confirms a weak negative correlation between price and sales.

	price	Sales Volume
price	1.00000	-0.33778
Sales Volume	-0.33778	1.00000



3.1- Promotion vs Sales volume:



	mean	median	max
Promotion			
No	872.377074	868.0	1262
Yes	1412.326540	1406.0	1940

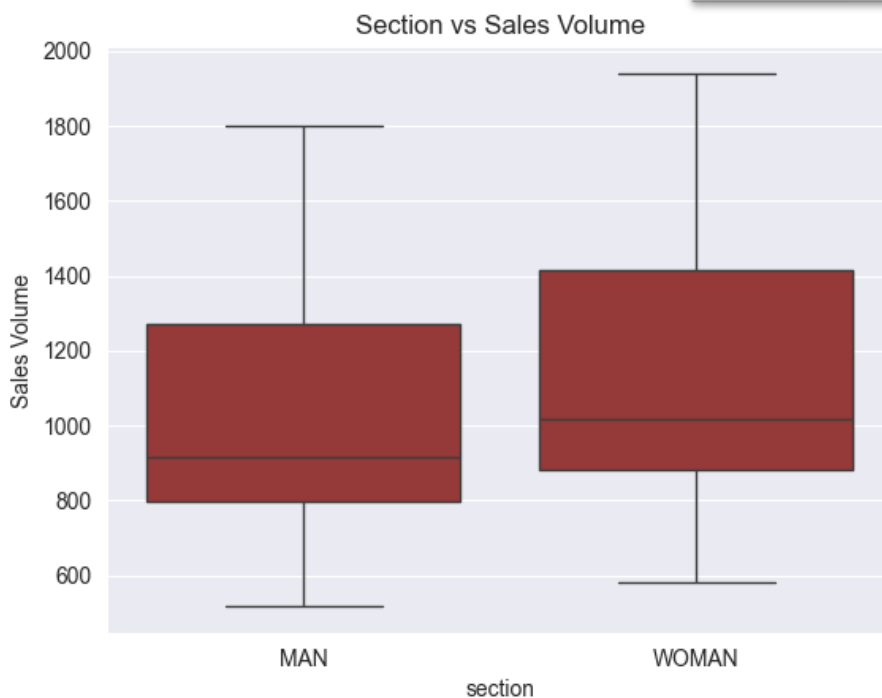
Products under promotion have a higher average and median sales volume than non-promoted products, indicating that promotions are effective in boosting sales.

The maximum sales are also higher for promoted products, showing that top-selling items are often on promotion.



3.3- Section vs Sales volume:

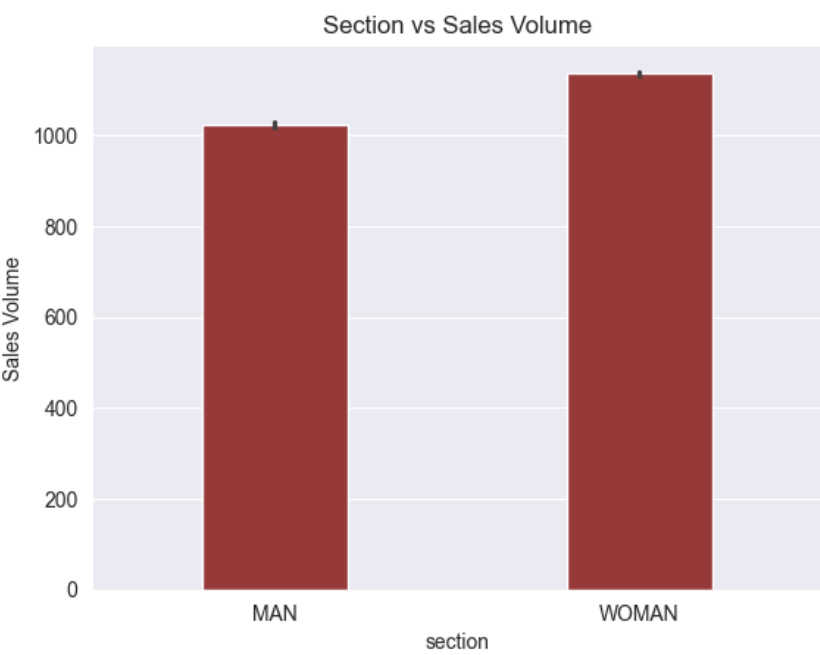
	mean	median	max
section			
MAN	1023.636325	914.0	1798
WOMAN	1136.347291	1018.0	1940



The analysis of sales by section shows that products in the Women’s section have higher sales volumes compared to the Men’s section.

This indicates that female customers contribute more to overall sales, and marketing or stock strategies could be adjusted accordingly.

Outliers in the Women’s section suggest a few highly popular products driving very high sales.

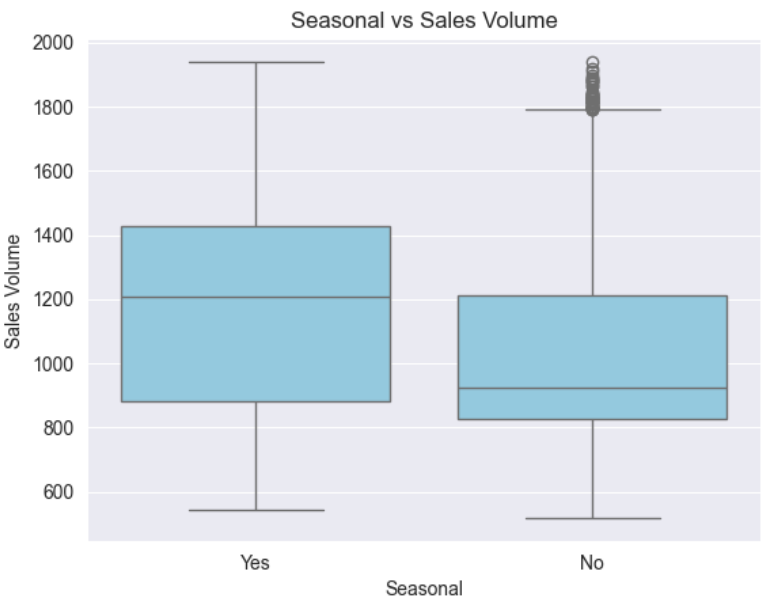


3.4- Seasonal vs Sales volume:

The analysis shows that seasonal products (Yes) tend to have slightly higher sales volumes than non-seasonal products (No).

This suggests that seasonal demand has a positive impact on sales, but the difference is moderate. Outliers indicate a few seasonal products performing exceptionally well

	mean	median	max
Seasonal			
No	1023.201164	926.0	1940
Yes	1171.746441	1206.0	1940



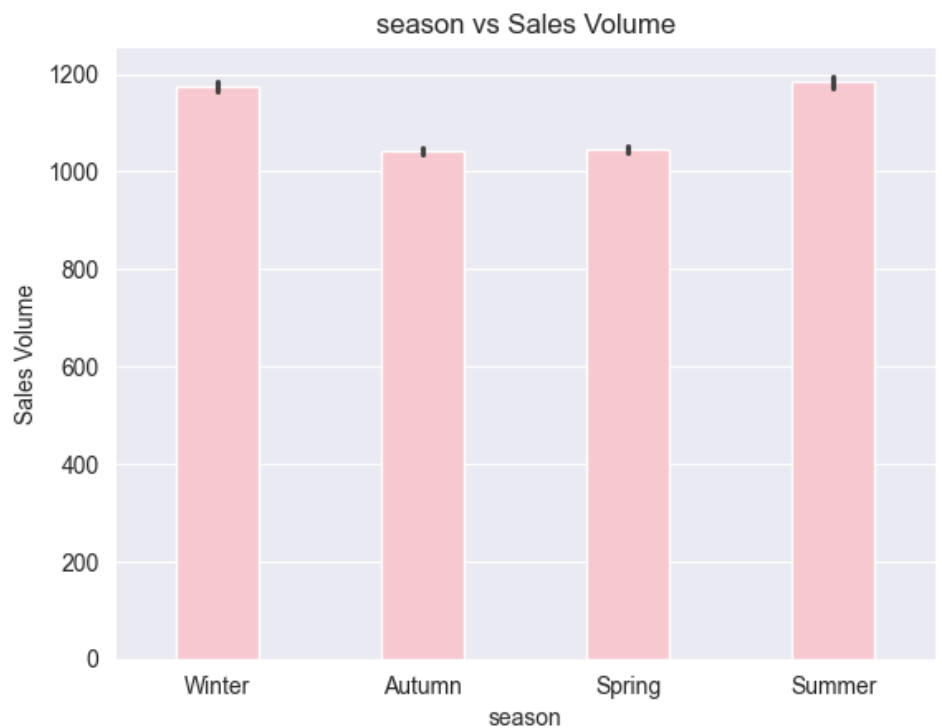
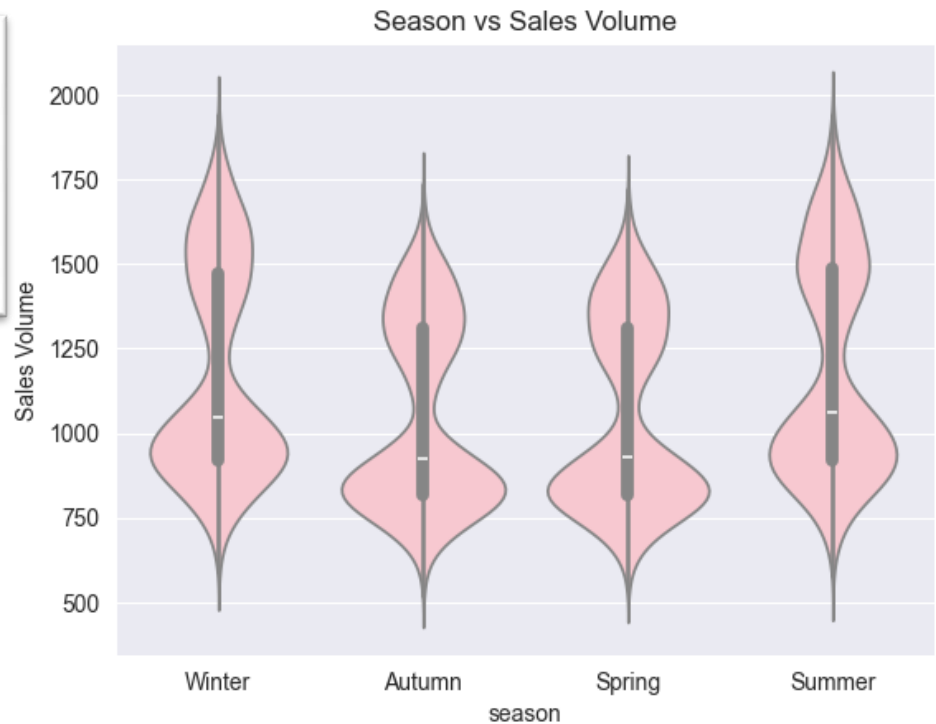
3.5- Season vs Sales volume:

	mean	median	max
season			
Autumn	1042.819830	926.0	1737
Spring	1046.223496	928.0	1719
Summer	1184.589470	1059.0	1940
Winter	1174.612558	1046.0	1940

Sales vary across seasons, with Summer and Winter showing the highest average and median sales.

This suggests that customer demand peaks in these seasons, and inventory or marketing strategies should focus more on Summer and Winter products.

Outliers indicate a few exceptionally high-selling products in these peak seasons.



3.6- Material vs Sales volume:

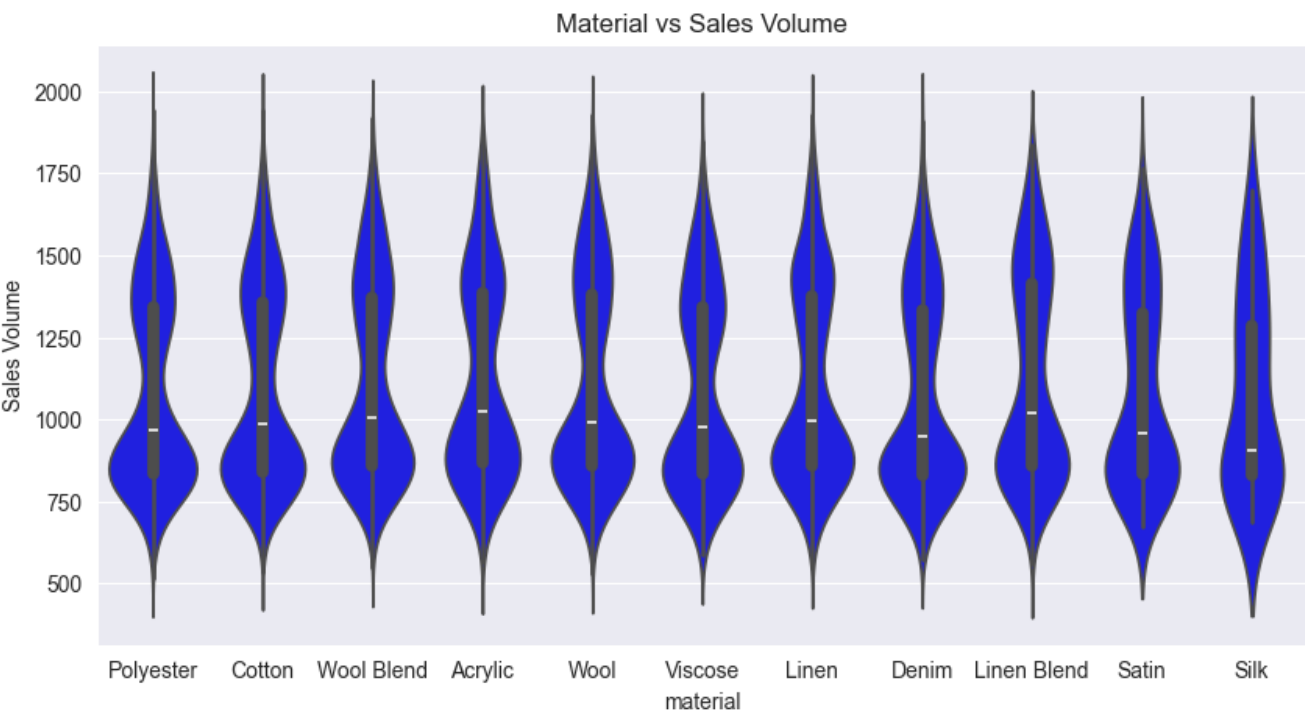
Sales performance varies across materials.

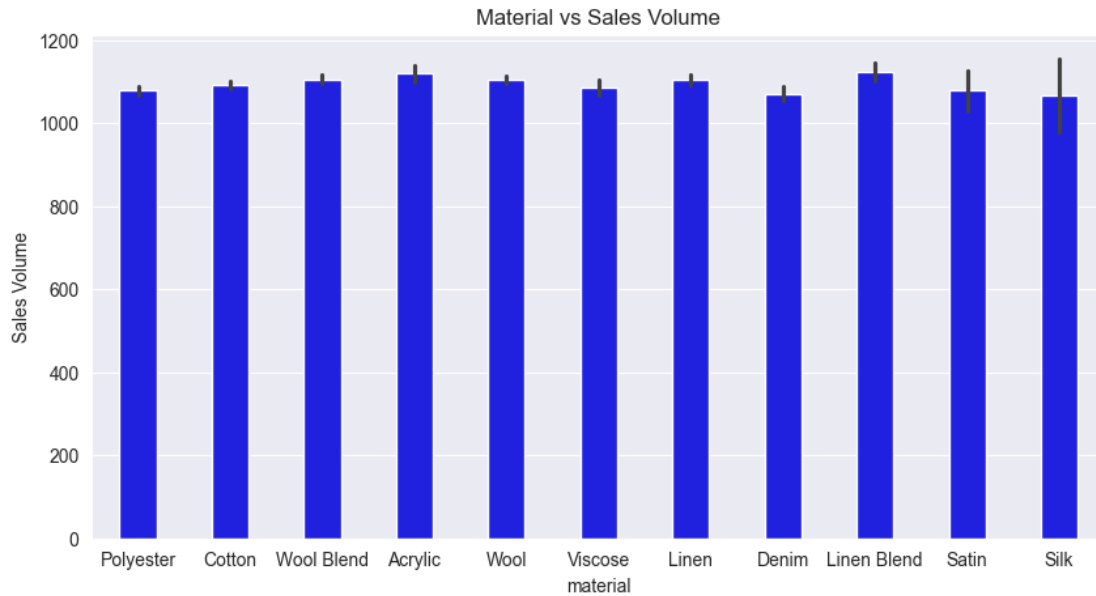
Some materials, such as Cotton and Wool, show higher average sales, indicating strong customer preference.

The difference between mean and median suggests the presence of outliers, where a few products achieve very high sales.

Additionally, materials with higher product counts tend to contribute more to total sales.

	mean	median	max
material			
Acrylic	1119.257662	1024.0	1864
Cotton	1091.226175	986.0	1940
Denim	1070.890944	947.0	1908
Linen	1104.127478	998.0	1926
Linen Blend	1125.240397	1021.0	1839
Polyester	1078.218018	968.0	1940
Satin	1078.666667	956.5	1767
Silk	1066.473684	906.5	1699
Viscose	1085.854545	978.5	1848
Wool	1105.521156	992.0	1929
Wool Blend	1106.111177	1004.0	1917

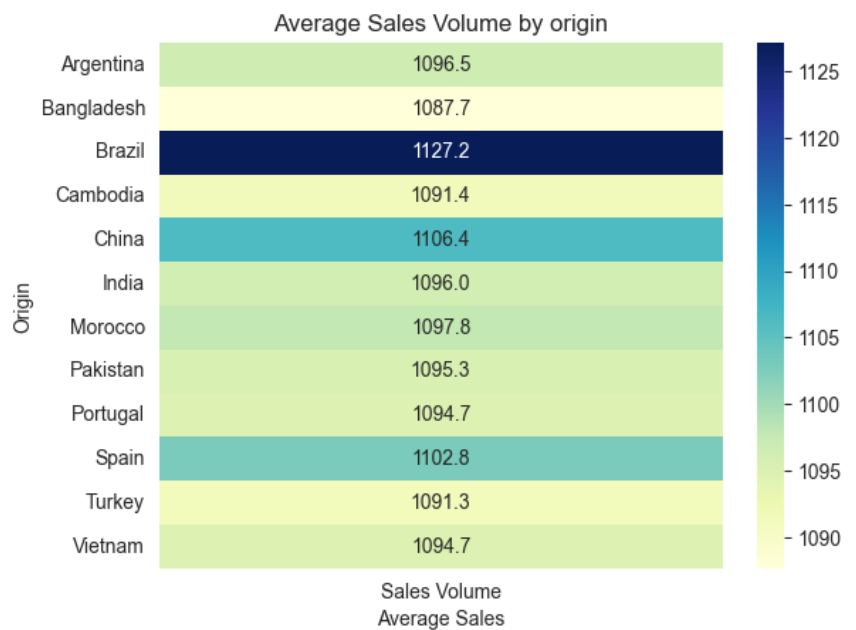


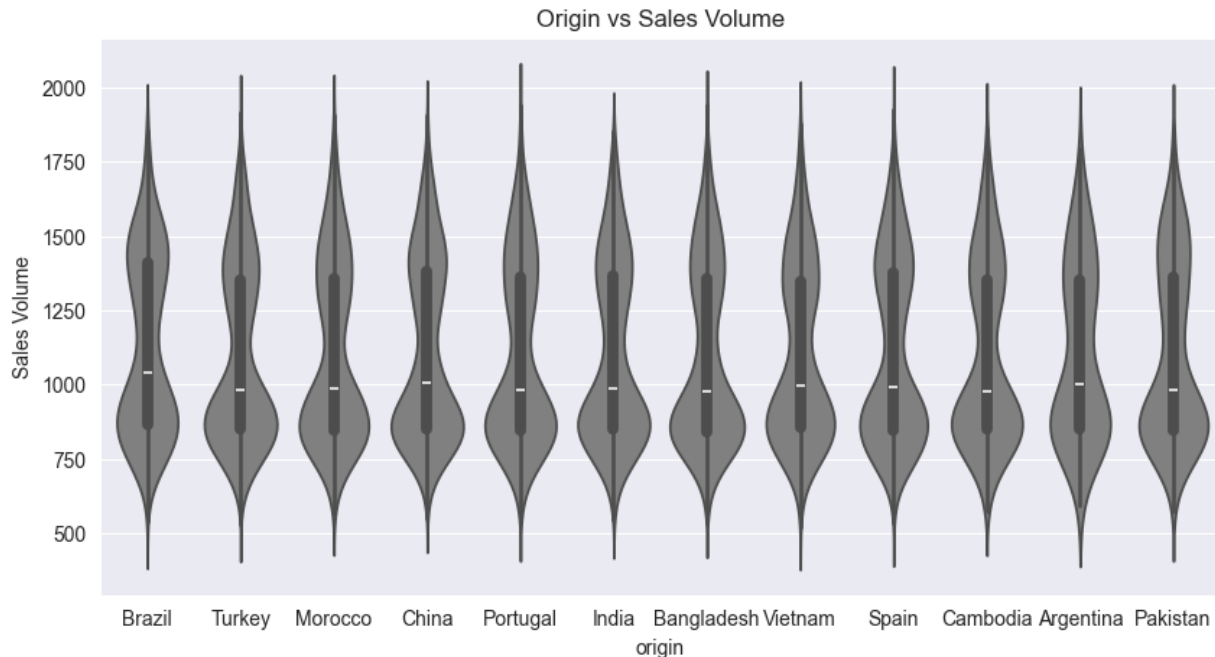


3.7- Origin vs Sales volume:

origin	mean	median	max
Argentina	1096.513966	1000.0	1795
Bangladesh	1087.657727	978.0	1940
Brazil	1127.154717	1040.0	1852
Cambodia	1091.422018	976.0	1864
China	1106.441133	1004.5	1908
India	1096.031972	987.0	1852
Morocco	1097.758621	989.0	1906
Pakistan	1095.285950	980.0	1844
Portugal	1094.742958	981.0	1940
Spain	1102.843750	992.5	1926
Turkey	1091.312727	984.0	1917
Vietnam	1094.718033	998.0	1878

The analysis shows that the average sales volume is relatively similar across all origins, indicating no significant difference in overall performance.





However, some countries such as Brazil, China, and Spain exhibit slightly higher average sales.

On the other hand, the maximum sales values vary significantly, with Bangladesh and Portugal achieving the highest peaks (1940 units).

This suggests that while overall performance is consistent, top-selling products are concentrated in specific origins, indicating the presence of strong outliers rather than uniformly high performance.

4-Key Insights & Business Implications:

- Promotions are mostly applied to lower-priced products.
- Women's section consistently shows higher sales volume than Men's .
- Seasonal trends: Summer and Winter are peak sales periods.
- Materials like Cotton and Wool dominate sales, suggesting consumer preference for natural fibers.
- Origin affects availability but not significantly the average sales.

Recommendations:

- Consider increasing promotions on mid-range products to maximize revenue.
- Focus marketing and inventory on Women's section during Autumn and Winter.
- Prioritize Cotton and Wool products for peak sales seasons

Final Analytical Summary:

- Dataset is clean and consistent, EDA shows strong trends and clear patterns.
- Key drivers of sales volume include Section, Season, Material, and Promotion strategy.
- Management can leverage these insights to optimize inventory, promotions, and marketing strategy.

Conclusion:

Zara's sales performance is influenced by product section, seasonal timing, material, and promotion strategy. Strategic planning around these factors can increase overall sales and improve product targeting.